

MODULE 9.3

Host-Based Intrusion Prevention

Cybersecurity Essentials

Topics Covered:

9.3.1 Host-Based Firewalls • Profiles • Logging • Distributed Firewalls

9.3.1 Firewall Examples • Windows Defender • iptables • nftables • TCP Wrappers

9.3.2 Host-Based Intrusion Detection (HIDS) • Architecture & Capabilities

9.3.3 HIDS Operation • Signature • Anomaly-Based • Policy-Based

9.3.4 HIDS Products • Cisco AMP • OSSEC • AlienVault • Tripwire

9.3.1 Host-Based Firewalls

Standalone software that controls traffic entering or leaving a single device

Host-based personal firewalls are standalone software programs that control traffic entering or leaving a computer. Firewall apps are also available for Android phones and tablets.

Predefined Policies / & Profiles

- Use profile-based rule sets to control packets
- Rules for Public, Private, and Domain networks
- Profiles are applied based on network type

Custom Rules

- Rules can be directly modified or created
- Control access by address, protocol, and port
- Granular inbound and outbound rule management

Alerts & User Control

- Issues alerts when suspicious behaviour is detected
- Offers the user a choice: allow or block the app
- Stops offending programs from running in future

Logging

- Records date, time, and allow/deny outcome
- Logs source/destination IP addresses and ports
- DNS lookups and routine events also appear
- Filtering tools needed to parse large log files

Distributed Firewalls

Centralised Management

A management server pushes firewall rules out to all hosts across the organisation — no manual config per device.

Log Aggregation

The management server also collects log files from every host, giving a single view of activity across all endpoints.

Best of Both Worlds

Combines the local protection of host-based firewalls with the consistency and oversight of network-based management.

9.3.1 Host-Based Firewall Examples

Four key tools — Windows and Linux solutions

Windows Defender Firewall

Profile-based firewall built into Windows (since XP).

Three profiles:

- PUBLIC — Most restrictive. Used on untrusted networks (coffee shops, airports)
- PRIVATE — For home networks isolated from the internet by a router/firewall
- DOMAIN — For trusted business networks with adequate security infrastructure

Supports logging and centralised management via Group Policy / System Center 2012.

iptables

Linux tool for configuring network access rules in the kernel's Netfilter modules:

- Rules are defined to ACCEPT, DROP, or REJECT packets
- Works at the kernel level — fast and powerful
- Applied to chains: INPUT, OUTPUT, FORWARD

The traditional and most widely-used Linux firewall tool.

nftables

The modern successor to iptables. Uses a simple virtual machine in the Linux kernel:

- Code runs inside the kernel VM to inspect network packets
- Implements decision rules for packet acceptance and forwarding
- Cleaner, unified syntax compared to iptables

Preferred on modern Linux distributions — replacing iptables over time.

TCP Wrappers

Rule-based access control and logging system for Linux:

- Filtering is based on IP addresses and network services
- Controls which remote hosts can access which services
- Logs connection attempts for auditing purposes

Lightweight and simple — works at the service level rather than packet level.

9.3.2 Host-Based Intrusion Detection (HIDS)

A comprehensive agent-based system that detects AND prevents malware

A HIDS is designed to protect hosts against known and unknown malware. It combines antimalware and firewall functionality — detecting threats AND preventing them from executing.

Log Analysis

Examines system and application log files for suspicious entries or patterns.

Event Correlation

Links related events across time and systems to identify attack sequences.

Integrity Checking

Verifies that critical files and system configs have not been modified.

Policy Enforcement

Ensures system behaviour complies with defined security policies.

Rootkit Detection

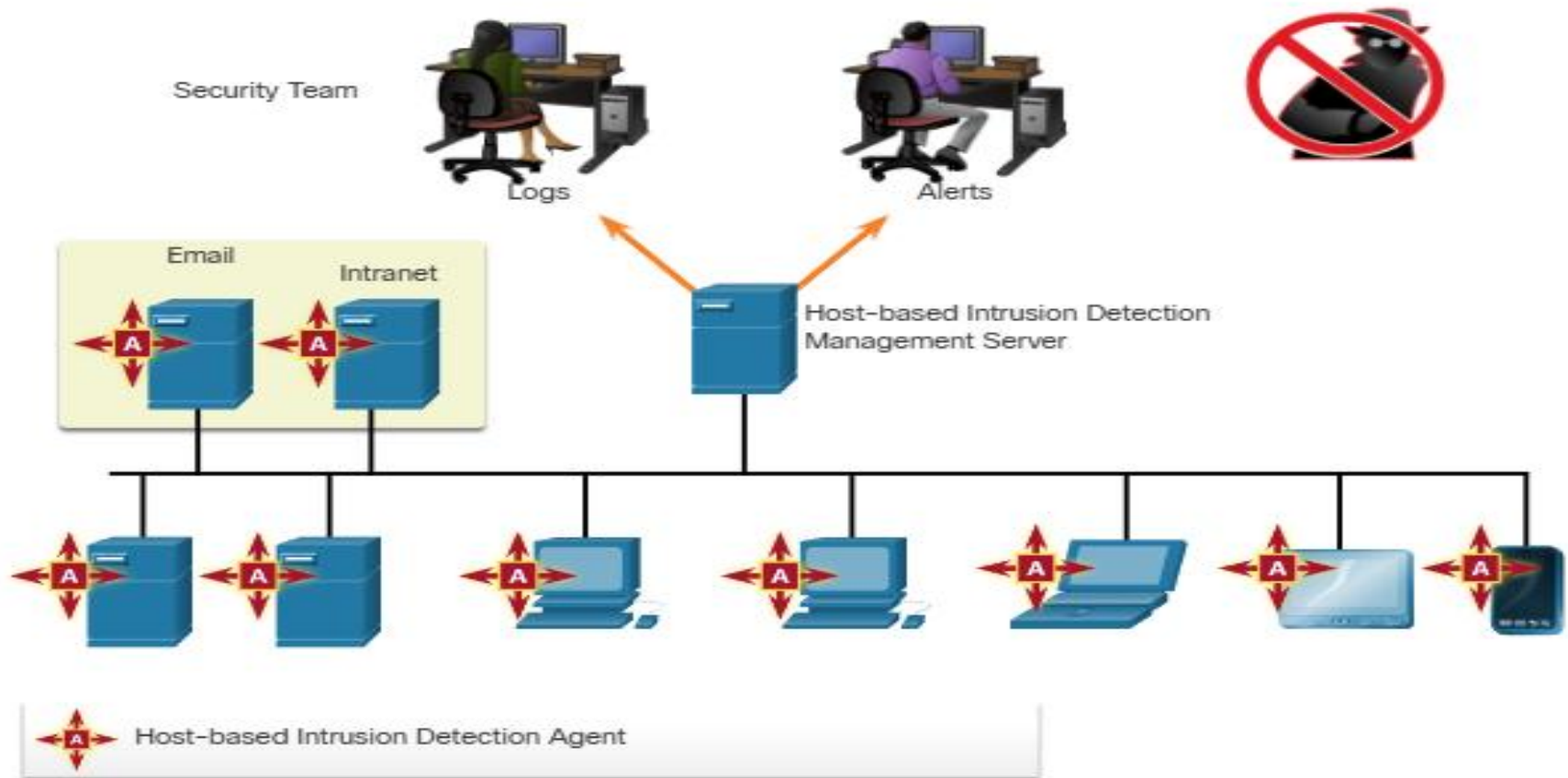
Scans for rootkits — malware that hides itself deep inside the OS.

Alerting

Sends alerts to security management systems and the security team in real time.

9.3.2 Host-Based Intrusion Detection (HIDS)

HIDS Architecture — How It Works:



9.3.3 HIDS Operation

Proactive and reactive strategies — three detection approaches

A HIDS functions as BOTH a detection AND prevention system. It uses proactive and reactive strategies, combining signature-based blocking with behavioural detection for unknown threats.

Layer 1 — Signature-Based Detection (Proactive)

Uses a database of known malware signatures to identify and BLOCK threats before they can execute. Fast and effective — but limited to known threats only. New (zero-day) malware and polymorphic malware (which constantly mutates its signature) can bypass this layer entirely.

Anomaly-Based Detection

Reactive — detects UNKNOWN threats

- Builds a baseline model of normal user and system behaviour
- Significant deviations from baseline are flagged as potential intrusions
- If intrusion is detected: logs details, sends alerts, and takes action
- Can cover zero-day and polymorphic malware that bypasses signatures

Limitation: Many non-malware events also change system behaviour — this can produce false positives, increasing workload and reducing trust in the system.

Policy-Based Detection

Reactive — enforces defined rules

- Normal behaviour is described by predefined rules
- Violation of rules triggers immediate HIDS action
- HIDS can shut down violating processes and log the event
- Alerts are sent to security personnel for review
- Admins can create custom policies distributed centrally

Most HIDS products ship with predefined rule sets. Custom policies can be pushed to all hosts from a central management system — no manual host-by-host config needed.

9.3.4 HIDS Products

Real-world tools that implement host-based intrusion detection and prevention

Most HIDS products combine host-based software agents with centralised security management — integrating with network security monitoring services and threat intelligence feeds.

Cisco AMP

- Advanced Malware Protection for endpoints
- Continuous monitoring and retrospective analysis
- Integrates with Cisco's Talos threat intelligence
- Detects, blocks, and responds to threats automatically
- Works across cloud, on-premise, and hybrid environments

AlienVault USM

- Unified Security Management platform
- Combines HIDS, SIEM, and vulnerability assessment
- Built-in threat intelligence from AlienVault OTX
- Centralised log management and event correlation
- Available as on-premise or cloud-based solution

Tripwire

- Specialises in file integrity monitoring (FIM)
- Detects unauthorized changes to critical system files
- Policy compliance and configuration assessment
- Real-time change detection and automated remediation
- Used widely in regulated industries (finance, healthcare)

OSSEC — (Open Source HIDS Security)

- Free and open-source HIDS
- Central Manager server + lightweight agents on hosts
- Agents available for Mac, Windows, Linux, Solaris
- Manager receives syslog alerts from network devices too
- Monitors system logs and checks file integrity
- Detects rootkits and can run scripts on event triggers

OSSEC is free and widely used — search 'OSSEC' to find documentation, community support, and download links.

9.3.5 Check Your Understanding

Key concept review — answers to the 4 quiz questions

Q1	What allows Linux system administrators to configure network access rules that are part of the Linux kernel Netfilter modules?	iptables	<i>iptables interfaces directly with the Linux kernel's Netfilter modules to define packet filtering and access control rules.</i>
Q2	Which of the following uses a profile-based approach to configuring firewall functionality?	Windows Defender Firewall	<i>Windows Defender Firewall uses Public, Private, and Domain profiles — each with different rule sets applied based on the network type.</i>
Q3	Which of the following is a rule-based access control and logging system for Linux?	TCP Wrappers	<i>TCP Wrappers provides rule-based access control filtered by IP address and network service, with built-in connection logging.</i>
Q4	Which of the following uses a learned baseline model to detect intrusions?	Anomaly-Based Strategy	<i>Anomaly-based detection builds a baseline of normal behaviour. Significant deviations from that baseline are flagged as potential intrusions.</i>

Module 9.3 — Summary

Host-Based Firewalls	Control traffic per device using predefined profiles and custom rules. Support logging and alerting.	Distributed Firewalls	Centralised management pushes rules to all hosts and collects logs — consistent policy across the org.
Windows Defender	Profile-based (Public / Private / Domain). Built into Windows. Centrally managed via Group Policy.	iptables / nftables	Linux kernel-level tools. iptables = traditional; nftables = modern successor using a kernel virtual
TCP Wrappers	Rule-based IP/service access control for Linux. Lightweight logging included.	HIDS	Agent-based system: log analysis, event correlation, integrity checking, rootkit detection, alerting.
HIDS Strategies	Signature (known threats) + Anomaly (baseline deviation) + Policy (rule violation). Three layers.	HIDS Products	Cisco AMP, AlienVault USM, Tripwire, OSSEC. All use agents + centralised management + threat intelligence.